KuaiRand-Pure
Week 1 Technical Report

Two-Tower Retrieval Baselines and Evaluation

SAR Recommendation Project

DRAFT – GPU acceptance pending

Abstract. This work implements a reproducible long-view retrieval baseline on the official Pure candidate pool. The system checks source identities, removes exact raw duplicates, rebuilds histories using observed playback completion, trains normalized user/item towers, exports aligned vectors and evaluates deterministic exact Top-100 retrieval. The report distinguishes measured results from pending acceptance evidence. Standard-exposure offline metrics describe held-out logged positives; they do not establish an online business improvement.

Current evidence. The corrected dataset contains 124,255 training positives and 30,589 validation-positive requests, against 7,583 candidate items. The train-popularity baseline is measured on this protocol. The earlier published enhanced checkpoint is retained as a separate historical diagnostic because its duplicate and history rules differ.

Acceptance status. The corrected basic/enhanced CUDA comparison and matching real-GPU interruption/recovery evidence are outstanding. This draft cannot certify Week 1 completion.

Generated from versioned experiment artifacts

1 Problem framing and supervision

At request time t, the system receives an anonymous user identifier, a scenario identifier and previously available positive feedback. It returns up to 100 videos from a time-eligible catalog. This is a **candidate-retrieval problem**: a user encoder and an item encoder map inputs into a common 64-dimensional space.

1.1 Target and unit of evaluation

The target is official `long_view` $= 1$ in standard, nonrandom exposure. For videos no longer than 18 seconds, the label requires watching at least the video's duration; for longer videos it requires at least 18 seconds. The label is copied from the official logs, not regenerated from our feature snapshot. One positive logged exposure is one evaluation request with one relevant item.

The learned score is normalized dot product,

$$s(u,i) = \langle f_\theta(u,t,h), g_\phi(i) \rangle, \qquad \|f_\theta\|_2 = \|g_\phi\|_2 = 1.$$

It is a ranking score, not a calibrated probability of long viewing. The model learns to rank a positive target above eligible comparison examples, rather than fitting an explicit positive/negative binary classifier on all exposures.

1.2 Why the baseline has three components

Popularity measures how far training-period positive frequency alone can go. The basic neural model uses user ID and request context with an ID-only item tower. The enhanced model adds a masked mean of positive history. Both neural runs share data, shuffle seed, output dimension and optimizer budget. Unversioned item content is excluded from the corrected comparison; the earlier content model is documented separately.

1.3 Interpretation boundary

Pure contains only logs involving the published candidate pool, so its histories omit other watched videos. Main validation uses standard exposure and can inherit its selection bias. Random exposure is kept separate, and the final test bundle remains sealed. The work does not claim unbiased online retention, CTR or revenue improvement.

2 Dataset and exploratory analysis

The source is the official KuaiRand-Pure archive. Its official MD5 is checked before processing. The shared Drive event/history/feature files are additionally matched to their frozen SHA-256 values. Raw event identities and primary fields are compared to the published event table before any new training artifacts are written.

Window	Exposures	Positives	Positive rate	Users	Items
warmup	886,326	297,920	33.61%	25,185	7,521
train	363,643	124,255	34.17%	25,769	6,882
validation	94,578	30,589	32.34%	20,565	5,279

Table 1: Retained nonrandom warmup/train/validation data after exact full-row deduplication. Counts are unique within each window; users/items can overlap across windows.

Training-period item positive counts have median 6, 95th percentile 101 and maximum 1,135. The top 100 items account for 26.83% of training positives. This concentration motivates the popularity comparator.

Training-user exposure counts have median 10 and 95th percentile 40. These are exposure counts, not the number of optimizer updates per user.

Observed video-duration quantiles (seconds) are: median 70.60, 75th percentile 139.43, and 95th percentile 288.60. The raw duration includes zero values; these are retained and reported, not replaced by a guessed duration.

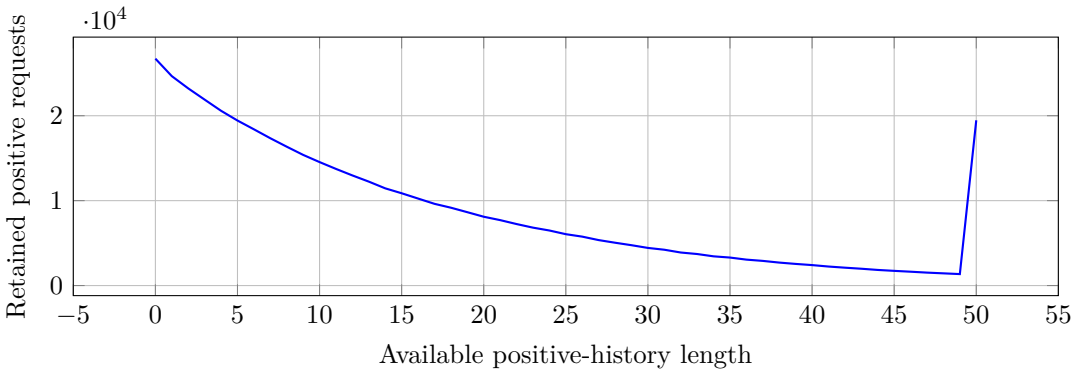


Figure 1: History-length distribution across retained warmup/train/validation positives; maximum length 50. Zero-history rate is 5.90%.

3 Preprocessing and temporal audit

3.1 Existing preparation

The publisher maps original anonymous IDs, timestamps, scenario, official label and random flag into standardized events. It assigns time splits and prepares static item metadata plus request-prior positive histories. The earlier consumer fits categorical dictionaries and numeric transformations from training-referenced items rather than the entire catalog.

3.2 Corrections introduced for this stage

The new audit removes 20,624 rows that are equal across *all* raw CSV columns, while retaining other repeated watches. Request IDs preserve the original filename and row number. No heuristic user-item deduplication is applied.

Histories now require

$$t_{\text{available}} = t_{\text{request}} + \texttt{play_time_ms} < t_{\text{current request}}.$$

The inequality is strict: a completion at the same millisecond is unavailable. A delayed completion is inserted in original request order, and the last 50 available positive observations are retained. The completion rule changes 1,038 histories relative to request-prior histories on the same deduplicated events. Random observations never become main-training histories.

This is a **conservative observed-playback proxy**. The source does not provide an independent feedback ingestion timestamp; therefore the audit does not claim exact production feedback availability.

3.3 Calendar consistency

Recorded date differs from timestamp-derived dates on 196,989 retained exposures under UTC and 11,235 under UTC+08. We do not silently declare either source calendar verified. Splits and periodic context instead use the explicit UTC timestamp protocol below.

Warmup	$t <$ 2022-04-15 00:00 UTC
Train	2022-04-15 00:00 $\le t <$ 2022-04-29 00:00 UTC
Validation	2022-04-29 00:00 $\le t <$ 2022-05-04 00:00 UTC
Test	Sealed; excluded from fitting and this report's EDA

4 User-side features and actual encodings

Feature	Source/meaning	Raw type	Encoded width
User ID	Anonymous training-window ID dictionary	int64	32
Scenario/tab	Request scenario; valid range 0–14	int64	8
Request time	Unix milliseconds; day/week cyclic context	int64	4
Positive history	At most 50 available prior item IDs	list[int64]	64 pooled
History mask	Valid history slots; padding is excluded	bool[50]	mask only

Table 2: User inputs. Basic omits history; enhanced includes its 64-dimensional masked mean.

User dictionaries are fitted to training-window users only and are numerically sorted. Known IDs map to indices $1, \ldots, V_u$; unseen users use index 0. This reserved embedding has no guaranteed learned cold-start behavior, so unknown-user coverage is reported explicitly. Tab values map to `tab` + 1; out-of-range scenarios use reserved index 0.

For timestamp T in milliseconds, with a zero UTC offset,

$$\alpha_d = 2\pi\,(T/1000 \bmod 86400)/86400,$$
$$\alpha_w = 2\pi\,((T/1000 + 3 \cdot 86400) \bmod 604800)/604800.$$

The four values are $(\sin\alpha_d, \cos\alpha_d, \sin\alpha_w, \cos\alpha_w)$. The three-day shift aligns the Unix epoch's Thursday with a Monday-based week. An actual retained timestamp, `1649475963278`, produces approximately

$$0.834017, \quad 0.551738, \quad -0.996517, \quad -0.083395.$$

Historical raw item IDs are left-padded with -1, distinct from real raw item ID 0. The shared item encoder generates $[B, 50, 64]$ vectors. The pooled history is

$$h_u = \frac{\sum_{\ell=1}^{50} m_\ell g(i_\ell)}{\max(1, \sum_\ell m_\ell)}.$$

An empty history therefore contributes a zero 64-vector and is kept in training/evaluation. Padding vectors never contribute through the mask.

5 Item metadata and feature decisions

Field	Meaning/unit	Actual examples	Values	Nulls	New use
item_id	Anonymous candidate ID; train-only embedding dictionary	0, 1	7583	0	ID embedding
author_id	Anonymous video author ID	7349781, 2103883	6510	0	Excluded
video_type	Published video type category	NORMAL, AD	3	0	Excluded
upload_type	Published upload channel/type	LongImport, Kmovie	14	0	Excluded
visible_status	Unversioned visibility snapshot	0.0	1	0	Excluded
music_id	Published music identifier	9155697141, 6355810746	7202	0	Excluded
music_type	Published music type code	9.0, 4.0	5	203	Excluded
tag	Published tag string (may contain multiple tags)	39, 2	110	96	Excluded
video_duration	Published duration snapshot, milliseconds	87433.0, 218066.0	5756	239	Excluded
server_width	Published video width, pixels	720.0, 960.0	156	0	Excluded
server_height	Published video height, pixels	1280.0, 720.0	120	0	Excluded
upload_dt	Upload date, day precision; eligibility proxy	2022-04-10, 2022-04-09	3	0	Eligibility only

The table describes all published item fields even when excluded from the corrected model. Cardinality and null counts are measured on the candidate metadata table, not on test responses. Numeric NaN values count as missing.

5.1 Corrected formal comparison

Both item towers use only `item_id`. Training-referenced item strings form a sorted dictionary; unseen-in-training candidate categories map to embedding index 0. An actual raw item ID `0` maps to index `1`. Unknown *candidate IDs outside the manifest* are rejected; an unseen training category within the valid candidate catalog is allowed.

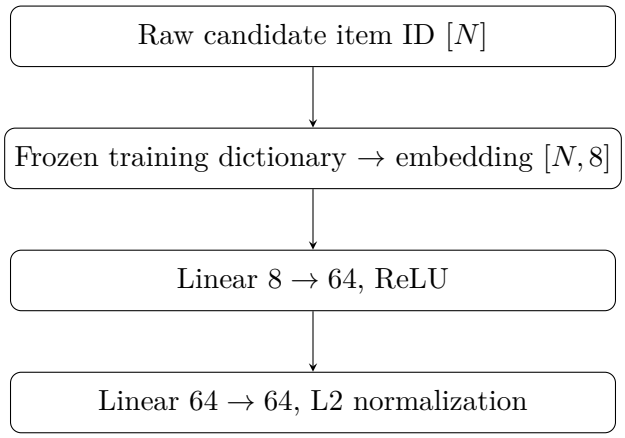
Snapshot metadata has no per-request historical version. Its availability cannot be established from this export, so author/type/music/tag/visibility/duration/size fields are excluded from new model inputs. Upload date remains only a conservative next-day candidate-eligibility proxy, with its limitations disclosed.

5.2 Historical model preprocessing

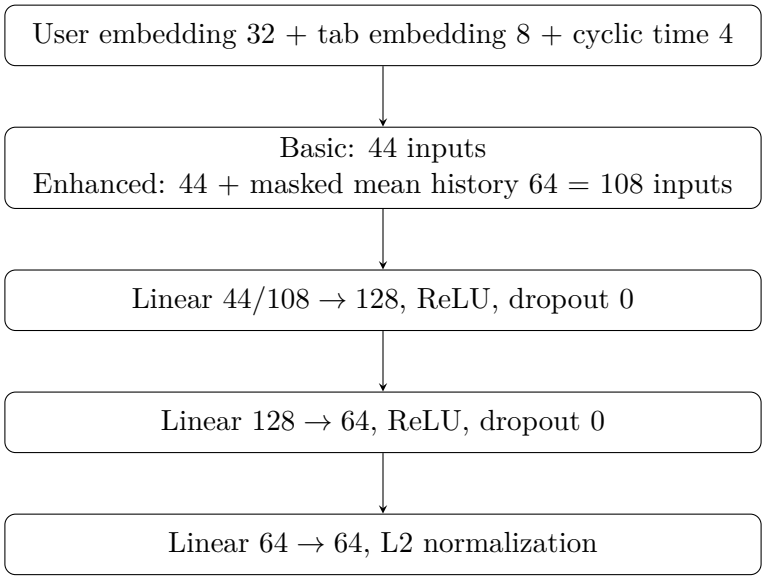
The old model uses eight categorical embeddings (8 dimensions each) and four normalized numbers: duration, width, height and relative upload day. It imputes training medians, subtracts training means and divides by training standard deviations; zero standard deviation becomes 1. Unknown categories use 0. Its original metadata buffers and configuration are preserved with the old checkpoint. This historical content model is not renamed as the corrected baseline.

6 Tower architectures

6.1 Item tower shared by both corrected models



6.2 User tower



Batch normalization is disabled. History and targets share item-tower parameters, so gradients from the user-history pathway and positive targets both update the same item encoder. The basic path does not encode unused histories. A singleton final batch is consumed rather than dropped.

The historical item MLP has input width $8 \times 8 + 4 = 68$ and layers $68 \to 64 \to 64$. Its history-enhanced user width remains 108, but its item representations and source protocol differ.

7 Loss, optimization and training budget

For a batch of B positive user-item pairs, let D_i contain off-diagonal targets sharing the same item ID as target i. Training uses

$$\mathcal{L} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(s(u_i, v_i)/\tau)}{\sum_{j \notin D_i} \exp(s(u_i, v_j)/\tau)}, \qquad \tau = 0.1.$$

The diagonal stays valid. Same-target off-diagonal entries are masked to $-\infty$ before cross-entropy. Different batch targets can still be false negatives: a user may also like them. No claim is made that these are observed dislikes. The initial baseline has no popularity/log-frequency correction for sampled comparisons.

Configuration	Corrected basic and enhanced
Output dimension	64, both towers normalized
Seed and shuffle	42; deterministic global row permutation
Optimizer	Adam, $\beta_1 = 0.9$, $\beta_2 = 0.999$
Learning rate / weight decay	0.001 / 0
Batch size / epochs	256 / 1 full positive-sample traversal
Precision / device	CUDA AMP for the acceptance runs
Dropout / batch normalization	0 / disabled
Temperature	0.1
History (enhanced only)	At most 50 available positive items, masked mean
Logging / checkpoint	Every 50 / 500 steps, plus epoch-end checkpoint

Table 4: Frozen Week 1 training defaults. There is no validation-based hyperparameter search in this one-epoch comparison.

Checkpoint state includes model parameters, Adam state, AMP scaler, Python, NumPy, PyTorch and CUDA RNG states, epoch, next-batch cursor and consumed count. Publication writes the model before a checksummed completion marker. Recovery selects only intact completed checkpoint files and validates binding of code, configuration, data, runtime and device.

Validation in-batch loss depends on batch composition and the available denominator. It is not interchangeable with Recall@100 or NDCG@100, and a smaller last batch can have a mechanically smaller loss.

8 Training results and provenance

Model	Consumed	Steps	Val. loss	Step+wait s	Peak GiB
basic	Pending	–	–	–	–
enhanced	Pending	–	–	–	–

Table 5: Corrected-protocol runs. Step+wait cost excludes checkpoint upload and full validation. Pending cells are not estimates.

The original run `b2ed5728b9a2` consumed 124,780 positives in 488 optimizer steps. Its validation in-batch loss is 5.516606. These are historical measurements under the original protocol. Local reproduction verified the frozen source, config, manifest, vector and checkpoint checksums before diagnostic inference.

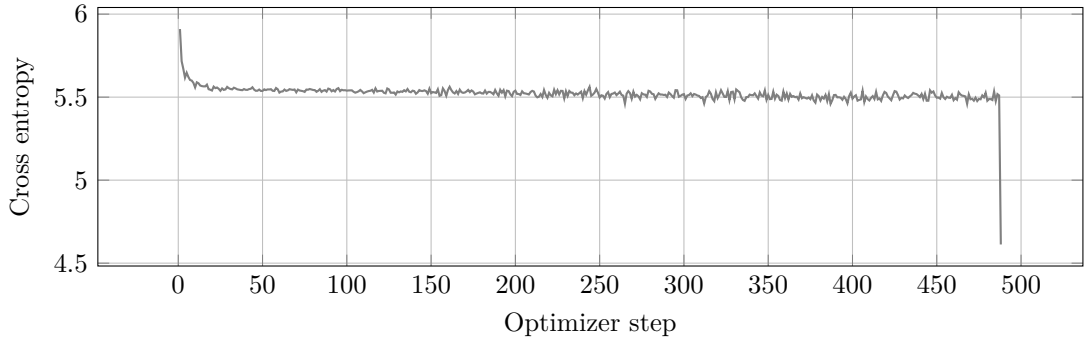


Figure 2: Historical enhanced run; the final batch has 108 examples rather than 256, explaining part of the last-step drop.

The private Drive retains full checkpoint and vector files; W&B stores run logs. Git retains source, checksums, aggregate evidence and this report. The source commit recorded by the notebook is immutable for a run; branch HEAD changes are not substituted for the actually executed version.

9 Evaluation protocol and measured results

Validation requests are all retained nonrandom positive exposures in the frozen validation window. The same request identities, catalog and eligibility rule apply to all three corrected-protocol systems. Available earlier validation feedback may enter later histories without updating model parameters. Repeat-view candidates are allowed.

Upload dates have day precision. An item becomes eligible at 00:00 UTC on the day after its upload date. Missing eligibility dates are excluded and counted. This is a conservative publication-date proxy; historical removals and visibility changes cannot be reconstructed from the snapshot.

Model	Requests	Recall@100	NDCG@100	Coverage	Short lists
popularity	30589	0.270784	0.069607	100.00%	0.00%
basic	Pending	–	–	–	–
enhanced	Pending	–	–	–	–

Table 6: Corrected-protocol exact retrieval. Unavailable targets are retained in the metric denominator.

For one relevant target per request, with rank r_q,

$$\text{Recall@100} = \text{HitRate@100} = \frac{1}{Q}\sum_q \mathbf{1}[r_q \le 100],$$

$$\text{NDCG@100} = \frac{1}{Q}\sum_q \frac{\mathbf{1}[r_q \le 100]}{\log_2(r_q+1)}.$$

A target absent from the eligible candidate set is a miss. Covered-only metrics are secondary diagnostics, not replacements for the end-to-end result. Empty slices use null rather than a fabricated zero.

Scores are computed by bounded request/item matrix blocks. Each block is merged into an exact Top-100 using score-descending, raw-ID-ascending sorting, including boundary ties. Missing result slots are explicit -1 sentinels. Vector shards must cover the full manifest catalog exactly, without duplicate IDs, and match the checkpoint's binding and unit norm requirement.

10 Coverage, performance and reliability

The corrected validation package contains 3.97% unknown-user requests and 1.49% empty-history requests. The measured candidate target coverage is 100.00%; no positive target is silently dropped for unsupported users or missing eligibility.

The local popularity exact-retrieval run uses batches of 256 requests after one warmup batch. Its total retrieval cost is 8.103 seconds, with batch P50 0.0675 seconds and batch P95 0.0725 seconds. These are observed local CPU batch timings, not production single-request latency. Hardware and timing scope are retained in evaluation JSON; CPU/GPU runs must not be treated as the same performance setting.

10.1 Historical checkpoint diagnostic

Frozen-source CPU diagnostic inference on the original 30,780 validation positives yields Recall@100 0.023717 and NDCG@100 0.006615. This is not part of the corrected comparison: the old model was trained with original duplicates, request-prior histories and snapshot content. It cannot certify the new protocol or be used to infer a fair improvement over the new popularity baseline.

10.2 Correctness and acceptance

Tests cover hand-calculated rank metrics, block/full equivalence, deterministic ties, temporal eligibility, uncovered targets, short lists, duplicate candidate IDs, completion-time histories, same-millisecond requests, unknown users and empty history. Model integration tests verify ID-only feature exclusion, normalized vectors, strict checkpoint binding and deterministic CPU recovery.

Real CUDA acceptance deliberately interrupts the enhanced run at step 25 and restores it. The final model, optimizer, scaler, RNG state and sample cursor must match the uninterrupted reference exactly. CPU correctness does not stand in for this hardware check. The release generator requires both full CUDA model runs plus recovery evidence tied to the exact enhanced checkpoint.

10.3 Outstanding evidence

basic, enhanced, real CUDA model runs, matching CUDA interruption/resume verification

11 Week 2 work and experiment discipline

Workstream	Next task	Acceptance evidence
Data	Publish audited 27K data and stable nested user subsets; investigate date semantics and metadata provenance	Source/row reconciliation, fixed subset hashes, no future history
User model	Compare no-history versus masked mean history with the shared trainer	Same budget, seed, requests and candidate protocol
Item model	Admit content only with defensible as-of provenance; compare ID-only versus content	Frozen train-only transformations and explicit feature availability
Joint experiments	Run basic, history-only, item-content-only and combined configurations	Four comparable validation results and cost measurements
Evaluation	Add long-tail, cold-start and eligibility coverage slices	Frozen definitions, per-slice counts, null for empty groups
Scaling	Estimate full traversal and encoding cost from measured throughput	Explicit GPU type, memory and data-wait measurements

Table 7: Week 2 priorities. ANN work follows a validated exact-retrieval reference.

Do not widen the search to multiple model depths, dimensions and losses simultaneously. Keep the full traversal budget fixed for the four principal comparisons. Use validation Recall@100 for later selection, retain NDCG@100 and coverage, and preserve all runs including non-improvements.

The final test bundle remains unused for training, tuning and stage selection. Candidate/request changes are versioned; results across changed protocols are not described as direct model improvements. Learned scores remain retrieval scores until a separate calibration or ranking task is established.

11.1 Known limitations

1. Recorded date disagrees with fixed UTC and UTC+08 calendars; date is not used for splitting.
2. No logged feedback-completion timestamp; full observed playback used as conservative proxy.
3. Metadata is an unversioned snapshot; content features excluded, upload date used only as a catalog eligibility proxy.
4. Pure histories omit videos outside the candidate pool; standard-exposure validation is not an unbiased online estimate.
5. One seed and one epoch provide a baseline, not a stability estimate or evidence of optimal convergence.
6. The item ID embedding cannot supply content generalization for unseen training items; the reserved unknown category is not a trained content representation.
7. Batch comparisons can contain false negatives, and the initial loss has no popularity correction.

Executed source commit: `cf9f15646934a24312b3a5303ca0785ac5944099`

A Feature examples and reproduction contract

Field	Actual values	Values	Min	Max
`user_user_id`	20853	26938	0	27,284
`user_tab`	1	15	0	14
`user_time_ms`	1649475963278	1269263	1,649,475,963,278	1,651,622,398,321
`long_view`	0	2	0	1
`play_time_ms`	1286	151346	0	964,774
`duration_ms`	289866	5737	0	1,177,720

Table 8: Actual retained warmup/train/validation event examples; anonymous public IDs are not personal identities. User-prefixed names disambiguate event fields from item metadata.

Training batches contain $[B]$ tensors for user ID, target item ID, time, scenario and official label; history item IDs and mask are $[B, 50]$. The formal ID-only item tower ignores standardized compatibility columns `category` and `duration`. Nonrandom positive filtering uses `long_view` and `is_random`; request ID and split are protocol fields rather than learned features.

Prepared manifest SHA-256
`0c35d9c9f483bca90f909cd8bcf5f7f1c6c64e6c3aeea411ea3617d0cc92b7fc`

Validation request identity hash
`c622789c33544e0122abb8d151ff22a50d8bd2db91a99589fddfba6e7db28efd`

Candidate artifact hash
`9b49e12466f88e9cbf0634b0f8e767104d17a7a4145b7bd730d2275d638c636d`

Official raw archive SHA-256
`c814bf6f3624c0cfae83c57de3df26b2ed206e5c57bab4c4dcbfabbabe20cbf0`

The team Drive separates raw data, versioned datasets, experiment runs and frozen releases by project and stage. The public notebook is `week1_baseline.ipynb`. It reads mounted prepared files directly. Optional copying to `/content` is a performance cache and does not re-download the source dataset.

Rebuild this report with `build_report.py`, providing the audit, matching manifest, stage release and real run-log root. Final generation refuses incomplete stages. Compilation uses an isolated temporary directory; auxiliary files are removed automatically. The standalone source includes its plot data and architecture diagrams.

References

[1] C. Gao et al. *KuaiRand: An Unbiased Sequential Recommendation Dataset with Randomly Exposed Videos*. CIKM 2022, pp. 3953–3957. https://arxiv.org/abs/2208.08696.

[2] KuaiRand project. Official dataset definitions and downloads. https://kuairand.com/.